

TACARLINE INJECTION 1MG/ML

DESCRIPTION: TACARLINE INJECTION 1MG/ML: Clear, colourless and odourless solution

COMPOSITION: Each ml contains Adrenaline Acid Tartrate 0.18% W/V

PHARMACODYNAMICS:

Adrenaline acts on both alpha and beta adrenergic receptors of tissues innervated by sympathetic nerves, except the sweat glands and arteries of the face. It is the most potent alpha receptor activator. Adrenaline stimulates the heart to increased output; raises the systolic blood pressure; lowers diastolic blood pressure; relaxes bronchial spasm and mobilises liver glycogen, resulting in hyperglycaemia and possibly glycosuria.

PHARMACOKINETICS:

The effects after SC administration are produced within 5 minutes, which is slower than from IM administration. Endogenous plasma concentration in normal subjects are in the range 30-160 pg/ml. Rapidly taken up by the heart, spleen, several glandular tissues, and adrenergic nerves; only metabolites are detectable in CFS; readily crosses the placenta; protein binding, about 50% bound to plasma proteins. Rapidly metabolized by oxidative deamination & O-methylation followed by reduction of glucuronic acid or sulphate conjugation.

INDICATION:

It is used in the emergency treatment of anaphylactic shock due to reaction from insect bites and drugs and for symptomatic relief of serum sickness, urticarial, pruritus, angioneurotic, oedema, respiratory distress due to bronchospasm, and relief of hypersensitivity reactions to drugs and other allergens.

RECOMMENDED DOSAGE:

Usual adult and adolescent dose:

Bronchodilator (**S.C.**) - Initially 200 to 500 mcg (0.2 to 0.5 mg) (base) repeated every 20 minutes to 4 hours as needed, the dosage being increased up to a maximum of 1 mg per dose, if necessary.

Anaphylactic reactions (**I.M. or S.C.**) - Initially 200 to 500 mcg (0.2 to 0.5 mg)(base) repeated every 10 to 15 minutes if necessary as needed, the dosage being increased up to a maximum of 1 mg per dose if necessary.

Vasopressor (anaphylactic shock) (**I.M. or S.C.**) - Initially 500 mcg (0.5 mg) (base) repeated every 5 minutes if necessary; may be followed by I.V. administration of 25 to 50 mcg (0.025 to 0.05 mg) every 5 to 15 minutes as needed, if inadequate response to I.M. or S.C. administration.

Or **I.V. administration** - 100 to 250 mcg (0.1 to 0.25 mg) (base) administered slowly. May be repeated every 5 to 15 minutes as needed or followed by an **I.V. infusion** at an initial rate of 1 mcg (0.001 mg) per minute, the rate being increased to 4 mcg (0.004 mg) per minute, if necessary.

Paediatric dose:

Bronchodilator; or anaphylactic reactions (**S.C.**) – 10 mcg (0.01 mg) (base) per kg of body weight or 300 mcg (0.3 mg) per square meter of body surface up to a maximum of 500 mcg (0.5 mg) per dose, repeated every 15 minutes for two doses, then every 4 hours as needed.

Vasopressor (anaphylactic shock) (**I.M. or S.C.**) – 10 mcg (0.01 mg) (base) per kg body weight, up to a maximum of 300 mcg (0.3 mg), repeated every 5 minutes if necessary. 10 mcg (0.01 mg) (base) as **I.V.** per kg of body weight every 5 to 15 minutes as needed, if an inadequate response to I.M. or S.C. administration.

Paediatric population

The following doses of Tacarline Injection 1 mg/ml (1:1000) are recommended:

Age	Dose
Over 12 years	0.5 mg IM (0.5ml 1mg/ml (1:1000) solution) 0.3 mg IM (0.3ml 1mg/ml (1:1000) solution) if the child is small or pre-pubertal)
6 - 12 years	0.3 mg IM (0.3ml 1mg/ml (1:1000) solution)
6 months - 6 years	0.15 mg IM (0.15ml 1mg/ml (1:1000) solution)
Under 6 months	0.01mg/kg IM (0.01ml/kg 1mg/ml (1:1000) solution)

If necessary, these doses may be repeated several times at 5 - 15 minutes intervals according to blood pressure, pulse and respiratory function.

Do not give Tacarline Injection 1mg/ml solution for injection intravenously. Intravenous administration of adrenaline for anaphylaxis requires the use of a 1:10,000 adrenaline solution.

ROUTE OF ADMINISTRATION: For i.v., i.m., s.c. & i.v infusion

CONTRAINDICATIONS:

- Known hypersensitivity to sympathomimetic amines
- Shock (other than anaphylactic shock)
- Cardiac dilatation and coronary insufficiency
- Narrow angle (congestive) glaucoma
- Organic brain damage
- During general anesthesia with halogenated hydrocarbons or cyclopropane
- With local anesthesia in fingers, toes, ears, nose or genitalia - there is a danger of vasoconstriction producing sloughing of tissues in these areas.
- Labor - it may delay the second stage by inhibiting spontaneous or oxytocin-induced contractions of the pregnant human uterus.

WARNINGS & PRECAUTIONS:

Other beta-agonist sympathomimetic: Allow sufficient time to elapse before or after administering another beta-agonist sympathomimetic agent to avoid additive effects.

Disease states: Use with extreme caution in the elderly, and in patients with cardiovascular disease, cerebrovascular insufficiency, diabetes, hypertension, chronic lung disease, angina pectoris or hyperthyroidism.

General anesthesia: Concurrent use with cyclopropane or halogenated hydrocarbon anesthetics may produce fatal ventricular arrhythmias.

Diabetic patients: A greater increase may be produced in heart rate, blood glucose, lactate, glycerol and free fatty acids when adrenaline is administered to diabetic patients with autonomic neuropathy than in diabetics without neuropathy.

Sodium metabisulfite: This product contains sodium metabisulfite, which may cause allergic reactions in susceptible individuals.

The possibility of an allergic reaction to sodium metabisulfite should be considered in asthmatic patients who show paradoxical worsening of their condition following use of the drug.

Gangrene: Inter-arterial administration must be avoided as marked vasoconstriction may result in gangrene.

Warning: this product contains Sodium Metabisulphite that may cause serious allergic type reactions in certain susceptible patients. Do not use if known to be hypersensitive to bisulphites.

INTERACTION WITH OTHER MEDICAMENTS:

Other sympathomimetic agents: Adrenaline should not be administered concomitantly with other sympathomimetic agents because of the possibility of additive effects and increased toxicity.

General anaesthetics: Administration of adrenaline in patients receiving cyclopropane or halogenated hydrocarbon general anaesthetics that increase cardiac irritability and seem to sensitise the myocardium to adrenaline, may result in arrhythmias including ventricular premature contractions, tachycardia or fibrillation.

Cardiovascular drugs: Adrenaline should not be used in patients receiving high dosage of other drugs, eg. digoxin, that can sensitise the heart to arrhythmias.

CNS and other drugs: Tricyclic antidepressants, some antidepressants, some antihistamines and thyroid hormones may potentiate the effects of adrenaline, especially on heart rhythm and rate.

Beta Blockers: The administration of adrenaline to patients receiving non-selective beta blockers (eg. propranolol) may result in severe hypotension, followed by a reflex bradycardia, due to stimulation of adrenergic receptors.

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: Adrenaline has been administered to a large number of pregnant women and women of childbearing age without any proven increase in the frequency of malformations or other direct or indirect harmful effects on the fetus having been observed.

However, the use of adrenaline during labor is contra-indicated because it may delay the second stage by inhibiting spontaneous or oxytocin-induced contractions of the pregnant human uterus.

Use in Lactation: Adrenaline is excreted in the breast milk. The use of adrenaline in breast-feeding women is therefore not recommended.

SIDE EFFECTS:

Adrenaline may cause reactions such as fear, anxiety, tenseness, restlessness, headache, tremor, weakness, dizziness, pallor, respiratory difficulty and palpitation. Over dosage or inadvertent IV injection of usual subcutaneous doses of adrenaline may cause hypertension. Cerebrovascular or other haemorrhage and hemiplegia may result, especially in geriatric patients. Inadvertent IV injection of adrenaline has also been reported to have caused convulsions, metabolic acidosis, and renal failure with anuria. Repeated injections of adrenaline can cause necrosis as a result of vascular constriction at the injection site. Prolonged use or over dosage of adrenaline can result in severe metabolic acidosis. Pulmonary oedema has been associated with excessive parenteral administration of adrenaline and following topical aerosol application. Gas gangrene which can be fatal has been reported following intramuscular injection of adrenaline into the buttock or thigh. This appears to have been due to Clostridium organisms on the skin being deposited into muscle tissue during injection, with the vasoconstrictor properties of adrenaline enhancing the effects of the infection. High doses may result in ventricular arrhythmias Rigidity and tremor may be exacerbated in patients with Parkinsonism Syncopal episodes have been reported in children Psychiatric disorders may be exacerbated.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Symptoms:

Initial hypertensive episode followed by hypotension. Cardiac arrhythmias, cerebral haemorrhage and pulmonary oedema.

Treatments:

Rapid digitalization with digoxin 1mg IV in divided doses.

Or immediate IV injection of quick acting alpha-adrenoceptor blocking agent (eg. 5-10 mg phentolamine mesylate) followed by a beta-adrenoceptor blocking agent (eg. 2.5 to 5mg propranolol). Rapidly acting vasodilators such as glyceryl trinitrate has also been used.

INCOMPATIBILITIES:

Adrenaline is rapidly denatured by oxidising agents and alkalis including sodium bicarbonate, halogens, nitrates, nitrites and salts of iron, copper and zinc. Adrenaline may be mixed with 0.9% Sodium Chloride injection but is incompatible with 5% sodium chloride injection. The stability of Adrenaline in 5% dextrose injection decreases when the pH is greater than 5.5.

STORAGE CONDITIONS:

Store below 25°C. Protect from light.

Keep out of reach of children. Jauhkan daripada kanak-kanak.

SHELF LIFE:

Please refer to outer package.

PACK SIZE:

Pack in 10's ampoules of 1ml in a box.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

Lot 2599, Jalan Seruling 59 Kawasan 3,

Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.

1500010834 N.2